



**College of Computing and Informatics
Department of Computer Science**

Course Syllabus

Course Title

Multimedia Programming and Design

Course Code
1501247

Spring 2025/2026

Course Syllabus

Course Title	Multimedia Programming and Design				
Course Code	1501247				
Credit Hours	3	Contact Hours	3		
Course Category	Program required				
Course Type	Compulsory				
Pre-requisite(s)	1411116 - Programming I OR 1411115 - Introduction to Computer Science I OR 0211115 - Introduction to Computer Science I OR 1501115 - Intro. to Computer Science I OR 1501116 - Programming I				
Co-requisite(s)					
Semester	Spring 2025/2026				
Instructor Name	Email	Classroom	Section	Session Type	Time
Isam Mashhour Hasan Al Jawarneh	ijawarneh@sharjah.ac.ae	C12-011	61	Lecture Undergraduate	TR 0930 - 1045

Course Description:

This course covers practical and advanced programming concepts in the context of multimedia software. Students will explore fundamental programming issues applied to the use and representation of sound, graphics, animation, and text. They will be introduced to the key ideas of event-driven programming and object-oriented programming using primarily Java as the programming language. They will work with sophisticated programming aids such as integrated development environments and/or large code libraries.

Course Objectives/Goals:

The goals of this Course are to:

- Gain fundamental knowledge of processing multimedia objects
- Know and use interactive multimedia programming and object oriented programming design and implementation
- Control and modify multimedia components in programming
- Work in teams to design and implement a multimedia programming project

Course Learning Outcomes:

By the end of successful completion of this Course, the student will be able to:

1. Apply the fundamental concepts and components concerned with the representation and processing of multimedia elements
2. Design, implement and test relatively simple interactive graphic programs
3. Use event-driven programming for obtaining interactive input from the user
4. Apply object-oriented programming techniques in your design and implementation
5. Manipulate graphics, images, sound, text, and movies
6. Design and implement a multimedia programming project in a team

Alignment of Course Student Learning Outcomes to Program Student Learning Outcomes

Program Learning Outcomes (PLOs)	Course Learning Outcomes (CLOs)
Identify and analyze user needs and to take them into account in the selection, creation, integration, evaluation, and administration of computing-based system.	- Apply object-oriented programming techniques in your design and implementation - Manipulate graphics, images, sound, text, and movies - Apply the fundamental concepts and components concerned with the representation and processing of multimedia elements - Design, implement and test relatively simple interactive graphic programs - Use event-driven programming for obtaining interactive input from the user
Analyze a complex computing problem and apply principles of computing and other relevant disciplines to identify solutions.	- Apply object-oriented programming techniques in your design and implementation
Design, implement, and evaluate a computing and multimedia-based solution to meet a given set of computing requirements in the context of the program's discipline.	- Manipulate graphics, images, sound, text, and movies
Analyze a complex computing problem and apply principles of computing and other relevant disciplines to identify solutions.	- Apply object-oriented programming techniques in your design and implementation - Apply the fundamental concepts and components concerned with the representation and processing of multimedia elements
Design, implement, and evaluate a computing and multimedia-based solution to meet a given set of computing requirements in the context of the program's discipline.	- Apply object-oriented programming techniques in your design and implementation - Manipulate graphics, images, sound, text, and movies - Design and implement a multimedia programming project in a team - Design, implement and test relatively simple interactive graphic programs - Use event-driven programming for obtaining interactive input from the user
Function effectively as a member or leader of a team engaged in activities appropriate to the program's discipline.	- Design and implement a multimedia programming project in a team

Weekly Distribution of Course Topics/Contents

Week	Topic	Comments*	CLOs
1	Introduction to Java		1,4
2	Introduction to Object Oriented Programming		1
3	Classes and Objects		2,3
4	Inheritance and polymorphism		2,3
5	Introduction to JavaFX		1,2,4,5

Week	Topic	Comments*	CLOs
6	Drawing shapes		2,5
7	Drawing shapes		2,5
8	Images		2,5
9	Layouts		2,3,4
10	Handling Events		3,4,6
11	Animations		3,4,5,6
12	Audio		3,4,5,6
13	Video		3,4,5,6
14	Game Programming		3,4,5,6
15	Game Programming		2,3,4,5,6
16	Final Exam		1,2,3,4,5

Students' Assessment:

Students are assessed as follows:

Assessment Tool(s)**	Date	Weight (%)
Project	Throughout the semester	15
Class Activity	Throughout the semester	10
Midterm Exam	Midterm Exam Period	20
Final Exam	Final Exam Period	45
Quizzes	Throughout the semester	10
Total		100

Course Outcome Assessment Plan:

Course LOs	Teaching/Learning Method(s)	Assessment Tool(s)	Performance Indicators	UoS competencies	Other competencies
Apply the fundamental concepts and components concerned with the representation and processing of multimedia elements	- Lectures - Exercises	- Exams - Class Activity - Quizzes	70 % of student achieve 70% of the CLO	- Critical and analytical thinking	
Design, implement and test relatively simple interactive graphic programs	- Lectures - Exercises	- Exams - Class Activity - Quizzes	70 % of student achieve 70% of the CLO	- Critical and analytical thinking - Problem solving	
Use event-driven programming for obtaining interactive input from the user	- Lectures - Exercises	- Exams - Class Activity - Quizzes	70 % of student achieve 70% of the CLO	- Critical and analytical thinking - Innovation / creative thinking	

Course LOs	Teaching/Learning Method(s)	Assessment Tool(s)	Performance Indicators	UoS competencies	Other competencies
				- Problem solving	
Apply object-oriented programming techniques in your design and implementation	- Lectures - Exercises	- Exams - Class Activity - Quizzes	70 % of student achieve 70% of the CLO	- Critical and analytical thinking - Innovation / creative thinking - Problem solving	
Manipulate graphics, images, sound, text, and movies	- Lectures - Exercises	- Exams - Class Activity - Quizzes	70 % of student achieve 70% of the CLO	- Critical and analytical thinking - Problem solving	
Design and implement a multimedia programming project in a team	- Lectures - Exercises	- Projects	70 % of student achieve 70% of the CLO	- Oral communication - Written communication - Teamwork or collaboration - Intellectual curiosity and lifelong learning	

Scope and Nature of Assessment Tools:

Scope and Nature of Assessment Tools

The aim of this section is to clarify what is required and how to submit, grade and assess each assessment tool:

Exams:

This assessment tool is composed of two exams, a midterm exam (20%), and a final exam (45%). The midterm exam covers three learning outcomes of the course and covers the materials of the first seven chapters. The final exam covers all course learning outcomes and all course materials with all chapters included.

Class Activity:

There will be several class activities. Each one has an equal weight in the computation of the course grade. Class Activity assignments will be distributed on the eLearning system.

Quizzes:

There will be several quizzes. Each one has an equal weight in the computation of the course grade.

Project:

The objective of the project is to gain experience in designing and developing multimedia application using object oriented design principles. Moreover, students are to acquire team-work skills, thus they will be asked to group up in teams (5-6 students per team depending on the number of students). The project will be built step-

by-step, where techniques and principles are to be incorporated as they are given in class. Projects will be demonstrated in class at the end of the semester.

Teaching and Learning Resources:

Textbook(s):

Y. Daniel Liang. (2017). Introduction to Java and Data Structures. 11th ed. Pearson.

Recommended Readings:

Malik, D. S. (2012). C++ Programming: From Problem Analysis to Program Design. 6th ed. Course Technology. ISBN:978-113362638.

Other Resources:

Eclipse (Eclipse Installer 2021-12 R)

<https://www.eclipse.org/downloads/packages/>

After launching the installation, select Eclipse IDE for Java Developers

Attendance policy:

Attendance is compulsory. A student missing 10% of the total allocated course hours will receive 1st warning notice and a student missing 15% will receive 2nd warning notice. A student missing 20% will be forced to withdraw (in accordance with the university regulations).

Plagiarism/Cheating:

Students are expected to do their own work. You are allowed to work on assignments in teams only if specified by the instructor. In other words, students are encouraged to communicate about general principles of the course, but all assigned homework must be done on an individual basis. The instructor is available to provide any assistance that you may need. Cheating is considered a serious offense by the university. You should be aware of the severe penalty for cheating (refer to the student code of conduct published in the university catalogue).

Grading rubrics:

Grading Rubric for Course Projects					
Scaled Score:		1/0	2	3	4
Criteria	Weight	Below Standard	Approaching Standard	At Standard	Exceeds Standard
Introduction	10%	No clear introduction, absent, or no evidence.	The introduction states the main topic but does not adequately preview the structure of the report.	The introduction states the main topic and previews the structure of the report.	The introduction is engaging, states the main topic and previews the structure of the report.
Data	20%	Data are inadequate and selected following the wrong procedure.	Data are selected but procedure of data selection is not followed.	Data selected following the standard procedure but not described in report.	Data are correctly selected following standard procedure and adequately described in report.
Analysis and	20%	Calculated wrong	Calculated correct	Calculated correct	Calculated correct

Grading Rubric for Course Projects					
Scaled Score:		1/0	2	3	4
Criteria	Weight	Below Standard	Approaching Standard	At Standard	Exceeds Standard
Reporting of Results (if applicable)		measures and analysis not relevant.	measures but analyses are not correct.	measures but analyses are not adequate.	measures and analyses are correct and provide more insights.
Authenticity and Citation*	20-30%	Some contents are paraphrased; plagiarism is most likely.	Some content is paraphrased; plagiarism is likely.	Some content is paraphrased and no suspicion of plagiarism.	All contents are original and well written.
Writing	20-30%	Errors in language that cause the content to be mostly incomprehensible.	Some errors in language that often inhibit comprehension.	Some errors in language that do not often inhibit comprehension.	Almost no error in language and well written.
Teamwork	10%	Team did not collaborate or communicate well.	Team did not collaborate or communicate well. A lack of mutual respect was frequently noted.	Team did collaborate or communicate well.	Team did collaborate or communicate well with high level mutual respect.

Grading Rubric for Course Assignments					
Scaled Score:		0/1	2	3	4
Criteria	Weight	Below Standard	Approaching Standard	At Standard	Exceeds Standard
Focus: Purpose	10%	No awareness	Shows limited awareness of purpose.	Shows awareness of purpose.	Purpose is clear.
Main idea	10%	No main idea	Vague sense of a main idea, weakly supported throughout the paper.	There is a main idea supported throughout most of the paper.	Clearly presents a main idea and supports it throughout the paper.
Organization: Overall	5%	No sense of organization	There is a sense of organization, although some of the organizational tools are used weakly or missing.	Good overall organization, includes the main organizational tools.	Well-planned and well-thought out. Includes title, introduction, statement of main idea, transitions, and conclusion.
Organization: Paragraphs	5%	Paragraphs lack clear ideas	Some paragraphs have clear ideas, support from examples may be missing and transitions are weak.	Most paragraphs have clear ideas, are supported with some examples and have transitions.	All paragraphs have clear ideas, are supported with examples and have smooth transitions.
Content	50-60%	Content is not sound	Content is sound and solid; ideas are present but not particularly developed or	Well-presented and argued; ideas are detailed, developed and supported with	Exceptionally well-presented and argued; ideas are detailed, well-

Grading Rubric for Course Assignments					
Scaled Score:		0/1	2	3	4
			supported; some evidence, but usually of a generalized nature.	evidence and details, mostly specific.	developed, supported with specific evidence and facts, as well as examples and specific details.
Research (if assignment includes a research component)	10%	The paper does not use adequate research or if it does, the sources are not integrated well. They are not cited correctly according to MLA style, nor listed correctly on the Works Cited page.	Sources support some claims made in the paper, but might not be integrated well within the paper's argument. There may be a few errors in MLA style.	Sources are well integrated and support the paper's claims. There may be occasional errors, but the sources and Works Cited conform to MLA style sheet.	Sources are exceptionally well-integrated and they support claims argued in the paper very effectively. Quotations and Works Cited conform to MLA style sheet.
Style: Sentence structure, word, choice, tone	5%	Sentences aren't clear	Sentences are generally clear word usage generally appropriate, details are somewhat adequate but may have awkward structure or unclear content; there may be patterns of punctuation errors.	Sentences are clear, word usage appropriate, details are enough but may lack variation; a few may be awkward and there may be a few punctuation errors.	Sentences are clear, word usage appropriate; details are enough and varied in pattern, from simple to complex, with excellent use of punctuation.
Grammar and Mechanics	5%	Continuous errors	Shows a pattern of errors in spelling, grammar, syntax and/or punctuation. Could also be a sign of lack of proof-reading.	A few errors in grammar, spelling, syntax and punctuation, but not many.	Excellent grammar, spelling, syntax and punctuation.

Adapted from: <http://www.aur.edu/wp-content/uploads/2010/12/University-Writing-Rubric>